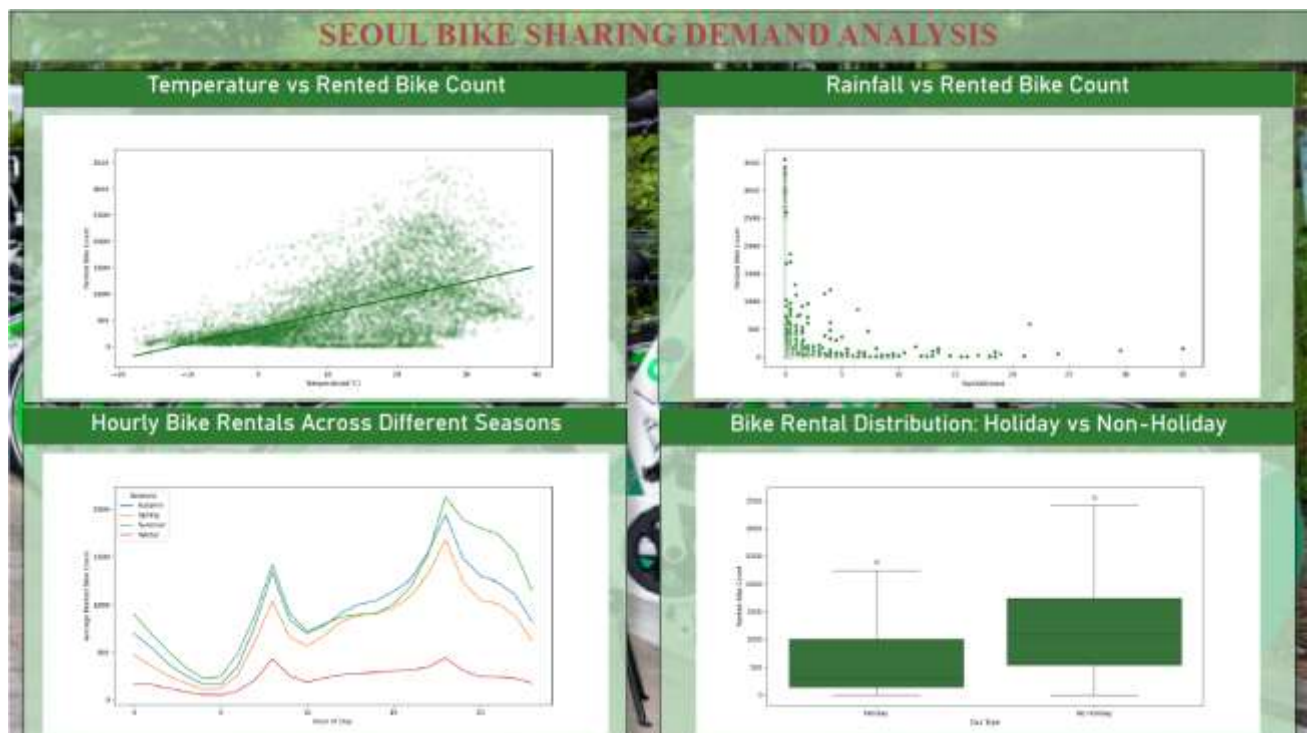


SEOUL BIKE SHARING DEMAND ANALYSIS

Power BI Dashboard



- Seasonal demand is highest in summer, while it drastically reduces during winter due to unfavourable weather conditions.
- Non-holidays have higher demand, proving bikes are used more for routine travel.
- Rentals drop to almost zero on non-functioning days, indicating that system availability directly influences the bike demand.

Business Recommendation:

1. Resource Optimization

Bike availability shall be increased during peak commute hours and summer months, and reduce supply during low-demand period.

2. Weather-Based Planning

The bike distribution and maintenance schedules can be adjusted based on weather forecasts, especially during heavy rain or snow.

3. Seasonal Maintenance

The major maintenance and repairs shall be planned during low-demand period like winter months.

4. Commuter-focused Policies

Weekly commuter passes or subscriptions can be introduced to target primary user base and improve customer retention.

5. System Reliability

Ensure maximum system uptime on working days, as even short periods of downtime can lead to a greater loss in demand.

Conclusion:

The dashboard clearly shows how bike rental demand works in Seoul.

Bike usage is highest during **morning and evening office hours** and increases in **warm, dry weather**.

Working days have much higher demand than holidays, showing that rentals are mainly driven by daily commuting.

Seasonal trends indicate **peak usage in summer** and **lowest demand in winter** due to weather conditions.

Overall, these insights help service providers **optimize bike availability, plan maintenance,** and support better **urban transportation planning**.